Species: It is the lowest taxonomic category. For example, the large Indian parakeet (*Psittacula eupatra*) and the green parrot (*Psittacula krameri*) are two different species of birds. They belong to different species *eupatra* and *krameri* and cannot interbreed.

Genus: It is a group of closely related species which constitute the next higher category called genus. For example, the Indian wolf (*Canis* pallipes) and the Indian jackal (*Canis aures*) are placed in the same genus *Canis.*

Family: A group of genera with several common characters form a family. For example, leopard, tiger and cat share some common characteristics and belong to the larger cat family *Felidae.*

Order: A number of related families having common characters are placed in an order. Monkeys, baboons, apes and Man although belong to different families, are placed in the same order Primates. Since all these animals possess some common features, they are placed in the same order.

Class: Related or similar orders together form a class. The orders of different animals like those of rabbit, rat, bats, whales, chimpanzee and human share some common features such as the presence of skin and mammary glands. Hence, they are placed in class Mammalia.

Phylum: Classes which are related with one another constitute a phylum. The classes of different animals like mammals, birds, reptiles, frogs and fishes constitute Phylum Chordata which have a notochord or back bone.

Kingdom: It is the highest category and the largest division to which microorganisms, plants and animals belong to. Each kingdom is fundamentally different from one another, but has the same fundamental characteristics in all organisms grouped under that Kingdom.

The taxa of living organisms are in a hierarchy of categories as follows.

Kingdom
Phylum
Class
Order
Family
Genus
Species

17.1.1 Basis for Classification

We can divide the Animal kingdom based on the level of organization (arrangement of cells), body symmetry, germ layers and nature of coelom.

Level of organization: Animals are grouped as unicellular or multicellular based on cell, tissue, organ and organ system level of organization

Symmetry: It is a plane of arrangement of body parts. Radial symmetry and bilateral symmetry are the two types of symmetry. In radial symmetry the body parts are arranged around the central axis. If the animal is cut through the central axis in any direction, it can be divided into similar halves. e.g. Hydra, jelly fish and star fish. In bilateral symmetry, the body parts are arranged along a central axis. If the animal is cut through the central axis, we get two identical halves e.g. Frog.

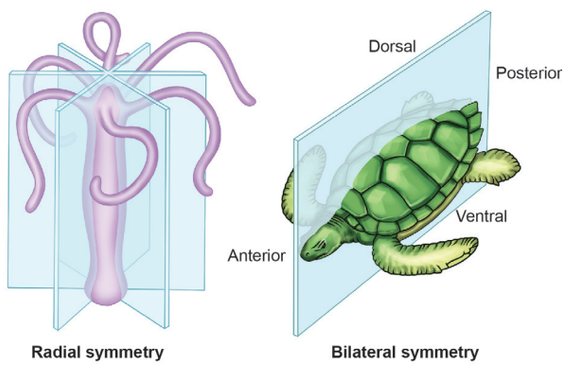


Figure 17.1 Radial and Bilateral Symmetry

Germ layers: Germ layers are formed during the development of an embryo. These layers give rise to different organs, as the embryo becomes an adult.

Organisms with two germ layers, the ectoderm and the endoderm are called

IX_SCI_EM_Unit-17_BIO.indd 200 12/7/2022 5:13:17 PM